

$$11. \begin{bmatrix} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix}$$

$$\text{Step 1: } \text{Row 2} = \text{Row 2} + 3 \times \text{Row 1}$$

$$\begin{bmatrix} 3 & -4 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix}$$

$$\text{Step 2: } \text{Interchange rows 2 and 3.}$$

$$\begin{bmatrix} 3 & -4 & 2 & 0 \\ -6 & 8 & -4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Step 3: } \text{Row 2} = \text{Row 2} + 2 \times \text{Row 1}$$

$$\begin{bmatrix} 3 & -4 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The corresponding equations are:

$$3x_1 - 4x_2 + 2x_3 = 0 \quad 0 = 0$$

$$\therefore x_1 = \frac{4x_2 - 2x_3}{3}$$

The general solution is

$$\begin{cases} x_1 = \frac{4x_2 - 2x_3}{3} \\ x_2 \text{ is free} \\ x_3 \text{ is free} \end{cases}$$

$$13. \begin{bmatrix} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Step 1: } \text{Row 1} = \text{Row 1} + \text{Row 3}$$

$$\begin{bmatrix} 1 & -3 & 0 & 0 & 9 & 2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Step 2: } \text{Row 1} = \text{Row 1} + 3 \times \text{Row 2}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -3 & 5 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The corresponding equations are:

$$x_1 - 3x_5 = 5 \quad (1) \Rightarrow x_1 = 5 + 3x_5$$

$$x_2 - 4x_5 = 1 \quad (2) \Rightarrow x_2 = 1 + 4x_5$$

$$x_4 + 9x_5 = 4 \quad (3) \Rightarrow x_4 = 4 - 9x_5$$

$$0 = 0 \quad (4)$$

The general solution is

$$\begin{cases} x_1 = 5 + 3x_5 & x_4 = 4 - 9x_5 \\ x_2 = 1 + 4x_5 & x_5 \text{ is free} \\ x_3 \text{ is free} \end{cases}$$